Run report: LEAD-19 projected-base protocol, card 1 (2026-10-23T21:54Z)

Observations only (research-visuals skill, "Every experiment run"). The interpretation is in `../../README.md` the protocol is `../../PROTOCOL.md`.

What ran

item	value
card	`cards/CARD-lead19-projected-base-20261023-1.yaml` (sha256 `3d655e8f…def9b`), compiled by `experiment-director`, run by `executor-mechanical` through `python3 tools/task_card.py run`
receipt	`cards/CARD-lead19-projected-base-20261023-1.receipt.json` (sha256 `bb2e3b71…5c86`), status `completed`, three commands, one attempt each, exit 0
commit	`6783523a3b75266db669d6138decdf142d14e16d` (the card's pin; the record's own `host.git_commit` is `null`, the binary was built outside a git-aware step)
binary	`target/release/ic`, sha256 `373c0f1d…e2ec` (bound by the card), blake3 `83689465…59b5` (recorded by the run)
command 2	`ic boundary --regime char2 --char2-degrees 15,18,21,24,27 --repeats 5 --seed 123212651130 --s4-max-degree 0 --base-arms raw,projected,subgroup --out research/lead19_projected_base/char2.json`, 36.9 s
command 3	`ic boundary --regime prime --prime-bits 24 --repeats 5 --seed 123212651130 --s4-max-degree 0 --base-arms raw,projected,subgroup --out research/lead19_projected_base/prime24.json`, 21.5 s
host	Intel Xeon @ 2.10 GHz (cloud container), 4 logical cores, linux x86_64
isolation	none earned and none needed: the metric is counted group operations and native counters; wall times are a practicality note
raw records	`../../char2.json` (1.48 MB), `../../prime24.json` (0.34 MB); stdout and stderr of both commands under `../../receipts/`
analysis	`../../analysis.json`, `../../analysis.md`, by `docs/ic/tools/lead19_projected_base_compare.py` (the frozen plan, PROTOCOL.md §8); recorded values only

Arms, instances, seeds

- Arms per instance and rung: `mitm_m2_negfold_walk` (A, the raw base), the same minus its class test (A′, derived: `total_gae` minus `admissibility_adds + admissibility_doubles`), `…_projected` (B), `…_subgroup` (C); each at the std rung and, where the ladder has one, the balanced rung (`…_balanced`). The ladder's other rows (`mitm_m3*`, `mitm_m2_negfold`) ran as always and are in the record but are not arms.
- Instances: `icv1-f2m15-t131-ed7256f0` (h = 2), `icv1-f2m18-t393-d2c04152` (h = 8), `icv1-f2m21-t2591-de54d72e` (h = 2), `icv1-f2m24-tm3047-2d0c0e8c` (h = 8), `icv1-f2m27-t12543-569dca8b` (h = 6); control `icv1-fp24-t1577-77336155` (h = 1).
- Seeds: curves fixed by `--seed 123212651130`; five targets per instance (repeats 0 to 2 are the frozen round-5 targets); eight matched-rho negation walks per target.
- Requested and completed: 5 instances × 5 repeats × 4 arms × 2 rungs = 200 arm rows on the char2 ladder, 1 × 5 × 3 × 2 = 30 on the control; all completed, none failed, none timed out.

Stop checks (PROTOCOL.md §6)

check	result
every row `verified`, both records `status: complete`	yes (`all_verified: true` in both)
prime-order control: `raw`, `projected`, `subgroup` bit-identical per repeat in `trials`, `relations_found`, `total_gae`, `rank`, every phase's `group_ops` and `native`	yes, no differences
arm A, repeats 0 to 2, equals the frozen round-5 counters (`trials`, `relations_found`, `total_gae`) on every char2 instance, both rungs	yes, no drift

Measured values

`S` = group-addition equivalents per √r, every phase, cold, priced with the repository's pinned ratios; mean over the five repeats, with every repeat's value in `analysis.json` (`s_per_repeat`). "rho" is the matched negation rho of the same process. The paired ratio is the geometric mean over repeats of S(A′)/S(B) with the 95 % percentile interval of 10,000 paired bootstrap resamples (seed 20261023). "yield" is pooled relations per trial.

row	h	rung	S(A)	S(A′)	S(B)	S(C)	S(A′)/S(B) (95 % CI)	S(A′)/S(C)	yield B/A	yield C/A	rho	S/rho: A′, B, C
f2m15	2	std (published)	27.65	15.78	15.79	12.49	1.00 (0.82, 1.21)	1.26	1.42	1.64	2.03	7.8×, 7.8×, 6.2×
f2m15	2	bal	26.40	26.40	20.71	18.16	1.24 (0.97, 1.50)	1.43	1.90	1.99	2.03	13.0×, 10.2×, 8.9×
f2m18	8	std	34.39	15.68	12.49	35.41	1.24 (1.08, 1.42)	0.44	3.28	3.01	1.96	8.0×, 6.4×, 18.0×
f2m18	8	bal	21.13	21.13	12.42	23.77	1.73 (1.29, 2.30)	0.88	3.72	3.68	1.96	10.8×, 6.3×, 12.1×
f2m21	2	std	23.11	15.22	9.49	11.15	1.58 (1.19, 2.16)	1.34	2.13	2.01	1.32	11.5×, 7.2×, 8.4×
f2m21	2	bal (published)	22.95	22.95	14.63	12.35	1.54 (1.35, 1.77)	1.90	1.74	1.88	1.32	17.4×, 11.1×, 9.3×
f2m24	8	std	36.38	25.74	16.68	27.89	1.53 (1.37, 1.68)	0.91	3.97	3.51	1.23	20.8×, 13.5×, 22.6×
f2m24	8	bal	25.59	25.59	11.19	15.58	2.37 (1.93, 3.12)	1.65	3.60	3.44	1.23	20.7×, 9.1×, 12.6×
f2m27	6	std	43.96	35.93	18.76	30.80	1.91 (1.70, 2.14)	1.16	5.46	5.77	1.08	33.3×, 17.4×, 28.5×

f2m27	6	bal (published)	41.47	41.47	13.76	17.75	2.85 (1.73, 4.99)	2.15	5.69	6.20	1.08	38.4×, 12.7×, 16.4×

Per-repeat `S` on the three primary rows (the spread behind the means):

row	rung	A′	B	C
f2m15	std	13.0, 16.7, 13.6, 17.7, 17.9	18.2, 16.1, 15.1, 12.6, 17.0	12.5, 9.9, 12.4, 13.8, 14.0
f2m15	bal	24.2, 37.1, 30.1, 24.2, 16.4	19.9, 22.4, 22.9, 17.6, 20.8	11.5, 21.3, 20.2, 18.0, 19.8
f2m21	std	18.0, 21.3, 11.9, 13.2, 11.7	9.5, 7.6, 7.9, 11.6, 10.9	13.4, 10.4, 9.0, 12.6, 10.3
f2m21	bal	16.9, 28.1, 28.2, 19.5, 22.1	13.0, 16.1, 14.5, 14.5, 15.1	14.1, 19.0, 9.2, 10.4, 9.0
f2m27	std	36.8, 30.3, 34.0, 40.5, 38.2	18.4, 18.8, 20.5, 18.6, 17.4	32.9, 31.7, 28.5, 29.2, 31.7
f2m27	bal	25.5, 59.5, 25.7, 29.2, 67.4	16.9, 7.5, 14.8, 13.1, 16.5	21.9, 18.0, 16.5, 18.5, 13.8

Trials and relations found, primary rows, published rung (per repeat):

row	arm	trials	relations found
f2m15 std	A	485, 621, 524, 746, 756	8, 12, 9, 10, 8
f2m15 std	B	797, 534, 649, 450, 661	13, 12, 13, 14, 14
f2m15 std	C	445, 250, 469, 550, 524	13, 5, 13, 11, 13
f2m21 bal	A	14160, 25479, 25557, 16863, 19395	16, 24, 28, 14, 20
f2m21 bal	B	10266, 13329, 11723, 11763, 12286	21, 19, 21, 23, 20
f2m21 bal	C	11350, 16340, 6405, 7730, 6215	20, 26, 15, 13, 17
f2m27 bal	A	100598, 259298, 101707, 117757, 295813	22, 71, 26, 32, 63
f2m27 bal	B	60226, 16886, 50307, 42154, 58325	82, 25, 67, 56, 87
f2m27 bal	C	59505, 41477, 34399, 43512, 21778	76, 68, 52, 77, 31

Class census of the folded pair tables (fraction of pairs with class sum zero; `classes` = distinct `[r]P` among the signed base points):

row	raw std	raw bal	projected (both rungs)	subgroup (both rungs)
f2m15	0.502 (2 classes)	0.510 (2)	1.0 (1)	1.0 (1)
f2m18	0.265 (4)	0.280 (4)	1.0 (1)	1.0 (1)
f2m21	0.501 (2)	0.505 (2)	1.0 (1)	1.0 (1)
f2m24	0.250 (4)	0.267 (4)	1.0 (1)	1.0 (1)
f2m27	0.167 (6)	0.169 (6)	1.0 (1)	1.0 (1)

Base-phase work the arms add (native counters of the arm bases, std rung):

row	B: `[h]P` doubles + adds (one scalar multiplication per abscissa)	C: abscissae scanned → lifted → kept, and the test
f2m15	15 + 0	69 → 32 → 15, 32 trace tests (unpriced)
f2m18	96 + 0	308 → 162 → 32, 162 `[r]P` tests = 2268 doubles + 1944 adds (charged)
f2m21	61 + 0	265 → 132 → 61, 132 trace tests (unpriced)
f2m24	411 + 0	1167 → 569 → 137, 569 `[r]P` tests = 11949 doubles + 3983 adds (charged)
f2m27	526 + 263	3119 → 1547 → 263, 1547 `[r]P` tests = 37128 doubles + 18564 adds (charged)

The class test A′ removes from A's std rung, in GAE per repeat (identical across repeats): 1516 (f2m15), 3384 (f2m18), 8068 (f2m21), 15400 (f2m24), 37992 (f2m27). The balanced rung never carried it.

Pricing band (PROTOCOL.md §8.5): the one unpriced native counter is arm C's trace test; at w = 0, 0.1 and 1 GAE per test, S(C) moves by at most 0.25 (f2m15 std, 12.49 → 12.74) and the A′ > B and A′ > C orders are unchanged on every row and rung (`order_holds_across_band: true` throughout).

Predictions (PROTOCOL.md §7), as the analysis script evaluated them:

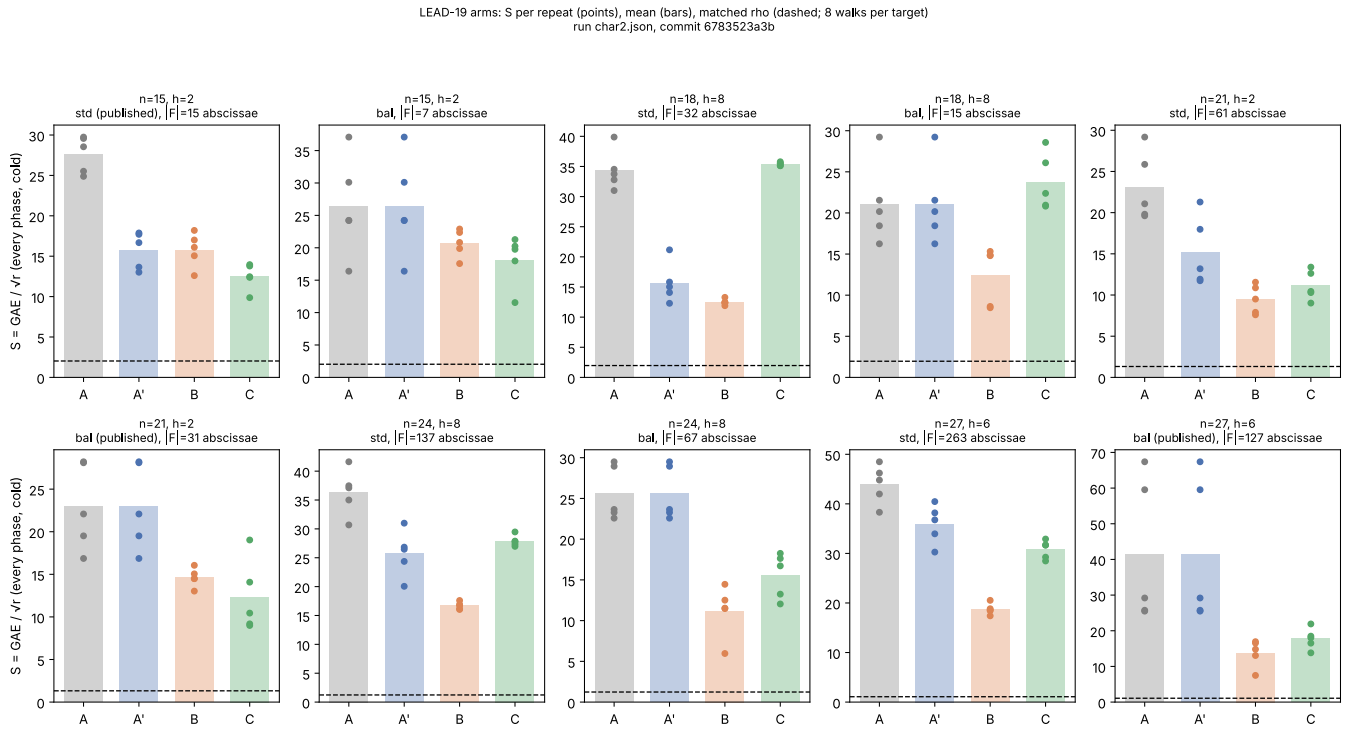
row	P1 class-zero fraction (predicted)	P2 yield B/A (band)	P3 S(A′)/S(B) at the published rung (predicted)	P4 yield B/C (h = 2)	P5 control
f2m15	0.502 (0.50): within 10 %	1.42 ([1.4, 2.6]): holds	1.00, CI (0.82, 1.21) (1.4–1.5)	0.87: within 15 %	identical
f2m21	0.501 (0.50): within 10 %	1.74 ([1.4, 2.6]): holds	1.54, CI (1.35, 1.77) (1.4–1.5)	0.92: within 15 %	identical

f2m27	0.167 (0.17): within 10 %	5.69 ([4.2, 7.8]): holds	2.85, CI (1.73, 4.99) (about 3)	not applicable	identical

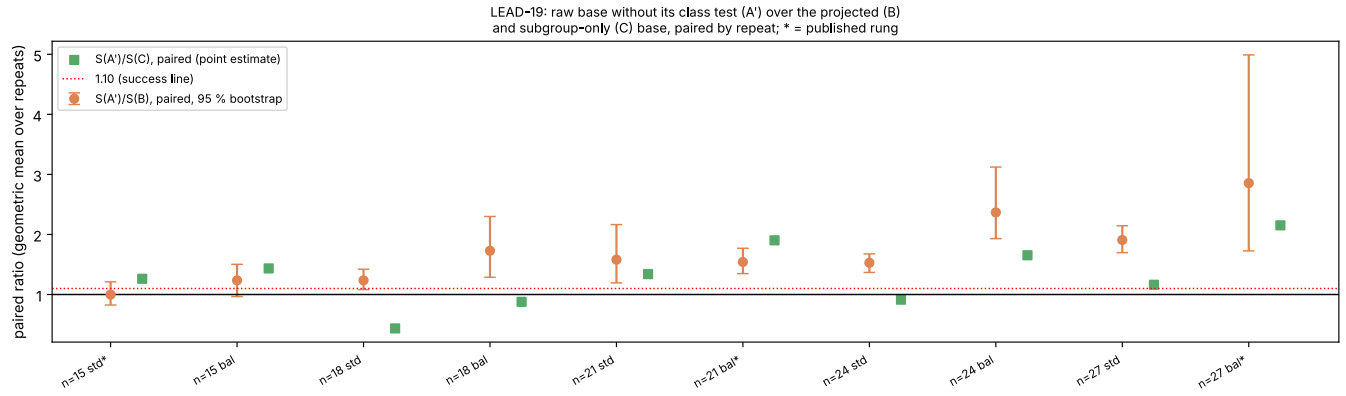
Published-row ratio (accounting, not the lever; frozen round-5 mean S over this run's mean): f2m15 26.31 / 15.79 = 1.67 (B), 2.11 (C); f2m21 24.39 / 14.63 = 1.67 (B), 1.98 (C); f2m27 36.93 / 13.76 = 2.68 (B), 2.08 (C).

The frozen comparison's verdict line (PROTOCOL.md §9, read on the published rung): **success on rows n = 21, n = 27** on n = 15 the interval straddles 1.10, which is the pre-registered `--repeats 20` extension.

Graphs



S per arm, rung and repeat; dashed line the matched rho



Paired ratios S(A′)/S(B) with the bootstrap interval and S(A′)/S(C), against the 1.10 line

Sources beside the graphs: `../../plot_lead19.py` reading `../../analysis.json`.

Missing or not measured

- No isolation tier, no `perf` counters, no wall-clock claim.
- No replay certificate, validator report or `VALIDATION.json` yet: this run is the lever's measurement, not a leaderboard row.
- The S_4 oracle was skipped (`--s4-max-degree 0`); it is not an arm.